

Beta Management Company

Read the case and answer the following questions.

A manager of a small investment company has been successfully using index funds for limited market timing. Growth has allowed her to move into picking stocks. She is considering two small and highly variable listed stocks, but is concerned about the risk that these investments might add to her “portfolio”.

I want us to think about the merits of the change of her investment strategy. This case provides a lead-in to the CAPM by learning about total risk, non-diversifiable or portfolio risk, and (CAPM) beta, and calculate variability of the stocks separately, and portfolio variance with and without the stocks, to see how an extremely risky (but low-beta) stock actually reduces risk. And, my hope is that we should be able to calculate stock betas.

These questions relate to the Beta Management Company case in your course packet. The data for this case is on the course website.

1. How risky are the three investments? Calculate the variability (standard deviation) of the stock returns of California REIT and Brown Group during the past 2 years. How variable are they compared with the Vanguard Index 500 Trust? Which stock appears to be riskiest? Based on the estimates of risk, which investment should have the highest expected return?
2. To analyze the effect of the asset allocation decision at Beta Management, let us examine three possible changes. Assume Beta Management currently has 100% of its assets in the Vanguard Index 500. Consider moving 1% of the assets to either (1.) a risk-free investment which pays 0.3% per month; (2.) California REIT; or (3.) Brown Group. Calculate the variability of this portfolio using each stock. How does each stock affect the variability of the equity investment, and which stock produces the riskiest portfolio? Explain how this makes sense in view of answer to Question (1).
3. Perform a regression of each stock’s monthly returns on the Index returns to compute the “beta” for each stock¹. How do these “betas” relate to answer to Question (2.)?
4. How might the expected return for each stock relate to answers in Question (1.) and (2.)?

¹ **Hint:** In this regression, the return of each stock will be the left hand side variable (or “y” variable) and the return on the index fund will be the right hand side variable (or “x” variable). The beta will be the estimated coefficient on x variable. If we are working in Excel, the functions SLOPE and LINEST will both estimate the beta. We should choose to have a constant in your regression – this is the default option for both commands.